



Vijay Academy Presents



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Data Science and Big Data Analytics (DSBDA)

TE Computer Engineering (2019 Pattern)

One Shot Lecture

Unit 3. Big Data Analytics Life Cycle

What You Get : Strategy For Exam :

- Simple Explanation
- PPT + Notes
- PYQs + Most IMP Questions



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Unit 3: Big Data Analytics Life Cycle



Syllabus Topics:

Introduction to Big Data

Sources of Big Data

Introduction to Data Analytics Life Cycle

Phase 1: Discovery

Phase 2: Data Preparation (Analytics Sandbox, ETLT Process)

Phase 3: Model Planning (Common Tools for Model Building)

Phase 4: Model Building

Phase 5: Communication Results

Phase 6: Operationalize

Extra Topics from PYQs:

Types of Analytics in Big Data (Descriptive, Diagnostic, Predictive)

BI vs Data Science

Data Driving Deluge

1: Introduction to Big Data

What is Big Data?

Big Data refers to extremely large and complex datasets that cannot be stored, processed or analyzed using traditional data processing tools and techniques.

Simply put — data that is too big, too fast or too complex for normal systems to handle.

Why did Big Data emerge?

Internet usage increased **massively**

Smartphones generate data every second

Social media produces **millions of posts** daily

IoT devices send continuous data

Online transactions happening every second

Traditional databases like MySQL **cannot handle** this much data



Why is Big Data Important?

Helps businesses make **better decisions**

Enables **personalized recommendations**

Improves **healthcare and research**

Enables **real time fraud detection**

Powers **artificial intelligence and ML**

Real Life Applications:

Netflix — recommends movies based on viewing history

Amazon — recommends products based on purchases

Google Maps — suggests routes based on real time traffic

Banks — detect fraudulent transactions instantly

Hospitals — predict patient health risks

Definition by Doug Laney (3 V's):

Big Data is defined by **3 V's**:

Volume — huge amount of data

Velocity — high speed of data generation

Variety — different types of data

Later extended to **5 V's** (Volume, Velocity, Variety, Veracity, Value)

2. Sources of Big Data

What are Sources of Big Data?

Sources of Big Data are the **places or origins from where huge amounts of data is generated continuously.**

Simply put — *where does all this big data come from?*

Structured vs Unstructured Sources:

Type	Sources	Example
Structured	Transaction data, Enterprise data	Sales records, Bank transactions
Unstructured	Social media, Media	Facebook posts, YouTube videos
Semi-structured	Web data, Emails	XML files, JSON data

Major Sources of Big Data:



1. Social Media

Platforms like Facebook, Instagram, Twitter, YouTube

Generates **millions of posts, likes, comments, shares** every second

Includes text, images, videos, audio

Example: 500 million tweets per day on Twitter

2. Machine and Sensor Data

Data from IoT devices, smart machines, sensors

Continuously generates real time data

Example: Smart watches tracking heart rate, GPS devices

3. Transaction Data

Data from business transactions

Online shopping, banking, payments

Example: Credit card transactions

4. Web and Internet Data

Data generated from website visits, clicks, searches

Example: Google processes 40,000 searches per second

5. Healthcare Data

Patient records, medical images, lab results

Electronic health records.

Example: MRI scans, X-rays

6. Communication Data

Emails, messages, calls, video conferences

Example: 300 billion emails sent per day worldwide

7. Public Data

Government data, weather data

Open datasets available publicly

Example: Climate data, traffic data

8. Enterprise Data

Data generated within organizations

ERP systems, CRM systems, supply chain data

Example: Company sales records, employee data, inventory data

3. Introduction to Data Analytics Life Cycle

What is Data Analytics Life Cycle?

Data Analytics Life Cycle is a **structured step by step process** that defines how data science and big data projects are executed from start to finish.

Simply put — *a roadmap that data scientists follow to solve a business problem using data.*

6 Phases of Data Analytics Life Cycle:



3. Introduction to Data Analytics Life Cycle



Phase 1: Discovery

Understand the business problem and define the goal of the project.
Identify available data sources and resources needed for the project.

Phase 2: Data Preparation

Collect, clean and preprocess raw data to make it ready for analysis.
Handle missing values, remove duplicates and transform data into required format.

Phase 3: Model Planning

Select appropriate analytical methods, algorithms and tools for building the model.
Explore relationships in data and decide which technique will best solve the problem.

Phase 4: Model Building

Actually build and train the analytical or machine learning model using prepared data.
Test and evaluate the model to check its accuracy and performance.

Phase 5: Communication of Results

Present the findings, insights and model results to business stakeholders.
Create visualizations and reports to explain results in a simple understandable way.

Phase 6: Operationalize

Deploy the final model into real world production environment.
Monitor model performance continuously and update it when needed.

4. Phase 1 – Discovery

Discovery is the **first phase** of Data Analytics Life Cycle where we **understand the business problem, available resources and data** before starting any analysis.

Simply put — *learning about the problem, the data and the resources before doing anything.*

Main Goal of Discovery Phase:

Understand **what problem** we are trying to solve

Identify **what data** is available

Understand **business objectives and requirements**

Assess **available resources** (people, technology, time)

Key Activities in Discovery Phase:



1. Learn the Business Domain

Understand the **industry and business context**

Learn **business terminology and processes**

Example: For a bank project, understand how banking works, what fraud means, etc.

2. Identify Business Problem

Clearly define **what question needs to be answered**

Define **goals and success criteria**

Example: "Why are customers leaving our app?" or "How to detect fraud?"

3. Identify Data Sources

Find out **what data is available**

Identify **internal and external data sources**

Example: Company database, social media data, government data

4: Assess Resources

Identify **team members and their skills**

Check available **technology and tools**

Estimate **time and budget** required

Example: Do we have enough data scientists? Do we have Hadoop cluster?

5. Frame the Analytical Question

Convert business problem into **analytical question**

Define **what type of analysis** is needed

Example: Business problem → "Sales are dropping"

Analytical question → "Which factors are causing sales to drop?"

6. Identify Potential Data Resources

Survey available data sources

Check **quality and quantity** of data

Example: We need 2 years of sales data but only have 6 months

5: Phase 2 - Data Preparation

What is Data Preparation Phase?

Data Preparation is the **second phase** where raw data is collected, cleaned and transformed into a format suitable for analysis and modeling.

Simply put — *making raw messy data clean and ready for analysis.*

Main Goal of Data Preparation Phase:

Collect data from **all identified sources**

Clean and **fix data quality issues**

Transform data into **required format**

Make data **ready for model planning and building**

Essential Python Libraries for Data Preparation:

Library	Use
Pandas	Data manipulation and cleaning
NumPy	Numerical operations
Matplotlib	Data visualization
Scikit-learn	Preprocessing and ML
Seaborn	Statistical visualization

Key Activities in Data Preparation Phase:



1. Data Collection

Gather data from **all identified sources**

Internal databases, external APIs, web scraping

Example: Collect customer data from CRM, sales data from ERP

2. Data Cleaning

Fix **missing values, duplicates, errors**

Example: Fill missing age values with mean, remove duplicate records

3. Data Integration

Combine data from **multiple sources** into one dataset

Example: Merge customer data from SQL database with sales data from Excel

4. Data Transformation

Convert data into **required format**

Normalize, standardize, encode categorical data

Example: Convert dates to same format,

5. Data Reduction

Reduce **size of dataset** while keeping important information

Remove irrelevant features

Example: Remove unnecessary columns, select only important features

6. Data Exploration

Explore data using **statistics and visualizations**

Understand **patterns, trends and relationships**

Example: Plot histograms, calculate mean and standard deviation

Analytics Sandbox :



Analytics Sandbox is a **separate isolated environment** where data scientists can freely explore, experiment and analyze data without affecting the main production systems.
Simply put — *a safe playground for data scientists to work with data freely.*

Why Do We Need Analytics Sandbox?

Data scientists need **freedom to experiment** with data
Cannot experiment on **live production systems** (too risky)
Sandbox provides **safe environment** to test ideas

What happens in Sandbox:

- Data exploration and cleaning
- Running experiments and algorithms
- Testing different models
- Visualizing data patterns
- Building prototype models

ETLT Process :

ETLT stands for **Extract** → **Transform** → **Load** → **Transform**

Simply put — a process of moving and converting data from source systems into analytics sandbox for analysis.



Each Step Explained:

E - Extract

Pull data from various source systems

Sources: databases, files, APIs, web

Example: Extract customer data from CRM database

T - Transform (First)

Clean and convert data into standard format

Handle missing values, fix errors, standardize formats

Example: Convert all dates to DD/MM/YYYY format

L - Load

Load transformed data into analytics sandbox

Store in HDFS, data warehouse or cloud storage

Example: Load cleaned data into Hadoop cluster

T - Transform (Second)

Further transform data for specific analysis needs

Apply domain specific transformations

Create new features, aggregate data

Example: Create new feature "customer age group" from age

6: Phase 3 - Model Planning

What is Model Planning Phase?

Model Planning is the **third phase** where data scientists decide **which analytical methods, algorithms and tools** will be used to build the model.

Simply put — *planning and deciding HOW we will solve the problem before actually building the model.*

Main Goal of Model Planning Phase:

Select **appropriate analytical techniques**

Identify **relationships in data**

Choose **right tools and technologies**

Create a **plan for model building**

Key Activities in Model Planning Phase:

1. Explore Data Relationships

Understand **relationships between variables**

Use statistical methods to find **correlations**

Example: Is there relationship between customer age and Exit rate?

2. Select Analytical Techniques

Choose **right algorithm** for the problem

Based on type of problem and data available

Example:

Prediction problem → Linear Regression

Classification problem → Logistic Regression,
Decision Tree

Grouping problem → K-Means Clustering

Association problem → Apriori Algorithm

3. Select Tools and Technologies

Choose **software and tools** for model building

Example: Python, R, SAS, MATLAB, Spark Mllib

4. Plan Model Building Workflow

Define **step by step plan** for building model

Decide **how to split data** into training and testing sets

Example: 70% data for training, 30% for testing

5. Check Data Availability

Verify if **enough data is available** for modeling

Identify if **more data is needed**

Example: Need at least 1000 records for reliable model

Common Tools for Model Planning:

Tool	Use
R	Statistical analysis and modeling
Python	General purpose ML and data science
SAS	Enterprise analytics
MATLAB	Mathematical modeling
SQL	Data querying and manipulation
Spark MLlib	Large scale ML on big data
Tableau	Data visualization

7: Phase 4 - Model Building

What is Model Building Phase?

Model Building is the **fourth phase** where data scientists **actually build, train and test** the analytical model using the prepared data and planned techniques.

Simply put — *actually creating and testing the model to solve the problem.*

Main Goal of Model Building Phase:

Build the actual model using selected algorithms

Train the model on prepared data

Test and evaluate model performance

Refine and improve model if needed



Key Activities in Model Building Phase:

1. Set Up Development Environment

Install and configure **tools and libraries**

Set up **computing environment**

Example: Set up Python with Scikit-learn, Jupyter Notebook

2. Split Data into Training and Testing Sets

Divide dataset into **training and testing portions**

Common split: **70-30 or 80-20**

Training set → used to **train the model**

Testing set → used to **evaluate the model**

3. Build the Model

Apply **selected algorithm** on training data

Model **learns patterns** from training data

Example: Train Logistic Regression model on customer data

4. Test the Model

Evaluate model on **testing data**

Check **accuracy, precision, recall**

Example: Test if model correctly predicts Customer Loss

5. Refine and Improve Model

Tune **model parameters** for better performance

Try **different algorithms** if needed

Example: If accuracy is 60% try different algorithm to improve it

6. Execute Model on Full Dataset

Run final model on **complete dataset**

Generate **predictions and insights**

8: Phase 5 - Communication of Results



What is Communication of Results Phase?

Communication of Results is the **fifth phase** where data scientists **present and explain** the findings, insights and model results to business stakeholders and decision makers.

Simply put — *telling the business what we found and what it means for them.*

Main Goal of Communication Phase:

Present **findings and insights** clearly

Explain results to **non technical stakeholders**

Help business make **data driven decisions**

Determine if results **meet business objectives**

Who are the Stakeholders:

Stakeholder

Business Executives

Project Sponsors

Business Users

IT Team

Data Scientists

What They Want to Know

High level insights and ROI

Did we achieve the goal?

How to use the results?

How to deploy the model?

Technical details and accuracy



Key Activities in Communication Phase:

1. Select Right Visualizations

Choose **appropriate charts and graphs**

Make data **easy to understand visually**

Example:

Bar chart for comparing categories

Line chart for trends over time

Pie chart for proportions

Scatter plot for relationships

2. Prepare Report and Presentation

Create **clear report**

Include **key findings and recommendations**

Use **simple language** for non technical audience

Example: PowerPoint presentation with key insights

3. Explain Model Results

Explain **what model predicts**

Highlight **important factors** that influence predictions

Example: "Age and usage frequency are top factors for customer loss"

4. Determine if Results Meet Objectives

Compare results with **original business goals**

Check if **success criteria** are met

Example: Goal was 80% accuracy — model achieved 88%

5. Identify Next Steps

Recommend **actions based on findings**

Suggest **improvements or further analysis**

Example: "Target customers aged 25-35 with retention offers"

9: Phase 6 – Operationalize

What is Operationalize Phase?

Operationalize is the **sixth and final phase** where the tested and approved model is **deployed into real world production environment** for actual use.

Simply put — *taking the model out of the lab and putting it into real world use.*

Main Goal of Operationalize Phase:

Deploy the model into production environment

Make model available for **real world use**

Monitor model performance continuously

Update and maintain model over time



Key Activities in Operationalize Phase:

1. Deploy the Model

Move model from **sandbox to production**

Integrate model into **existing business systems**

Example: Integrate loss prediction model into CRM system

2. Run Pilot Program

Deploy model on **small scale first**

Test in **real world conditions**

Check if model performs well in production

Example: Test loss model on one region before full rollout

3. Finalize Code and Documentation

Clean and **finalize the code**

Write **technical documentation**

Example: Document how to use loss prediction dashboard

4. Monitor Model Performance

Continuously **track model accuracy**

Example: Check monthly if loss prediction accuracy is maintained

5. Retrain and Update Model

Retrain model with **new data** periodically

Update model when **business conditions change**

Example: Retrain model every 3 months with latest customer data

6. Measure Business Impact

Measure **business value** of model

Check if model is **helping achieve business goals**

Example: Did loss rate reduce after implementing model?



10: Types of Analytics in Big Data

What are Types of Analytics?

Analytics in Big Data refers to **different ways of analyzing data** to get insights and make decisions.

There are **4 types** of analytics:

1. Descriptive Analytics

Describes **what happened** in the past using historical data.

Simply put — *looking at past data to understand what occurred.*

Question answered: "What happened?"

Techniques: Summary statistics, reports, dashboards

Tools: Tableau, Power BI, Excel

Example: Monthly sales report showing total revenue last year

2. Diagnostic Analytics

Explains **why something happened** by finding causes and relationships.

Simply put — *digging deeper to find the reason behind what happened.*

Question answered: "Why did it happen?"

Techniques: Drill down analysis, correlation, data mining

Tools: SQL, Python, R

Example: Finding why sales dropped in December by analyzing customer complaints and market trends

3. Predictive Analytics

Predicts **what will happen** in the future using statistical models and ML.

Simply put — *using past data to forecast future outcomes.*

Question answered: "What will happen?"

Techniques: Regression, Decision Trees, ML algorithms

Tools: Python Scikit-learn, SAS, R

Example: Predicting which customers will leave next month

4. Prescriptive Analytics

Recommends **what action to take** to achieve desired outcome.

Simply put — *not just predicting future but also suggesting best action.*

Question answered: "What should we do?"

Techniques: Optimization, simulation, AI

Tools: IBM SPSS, SAS, Python

Example: Recommending which customers to target with discount offers to prevent churn

11: Data Driving Deluge

Data Driving Deluge refers to the **massive flood or explosion of data** being generated

every day from various sources across the world.

Simply put — *the world is drowning in data and it keeps growing at uncontrollable rate.*

What is Driving this Data Deluge?

1. Internet Growth

Billions of people connected to internet daily

Every click, search and visit generates data

Example: 5 billion people use internet daily

2. Social Media Explosion

Millions of posts, likes, shares every second

Example: 500 million tweets per day

3. Smartphone Revolution

Every smartphone generates location, usage, communication data

Example: 6.8 billion smartphone users worldwide

4. IoT Devices

Smart devices continuously send data

Example: Smart watches, home automation, industrial sensors

5. Cloud Computing

Easy and cheap storage leads to more data being saved

Example: AWS, Google Cloud, Azure storing petabytes of data

6. E-Commerce Growth

Every transaction, click and purchase generates data

Example: Amazon processes millions of orders daily



BI vs Data Science :

Difference Table:

Point	Business Intelligence	Data Science
Focus	Past and present data	Future predictions
Question	What happened?	What will happen?
Tools	Tableau, Power BI, Excel	Python, R, ML algorithms
Users	Business managers	Data scientists
Output	Reports, dashboards	Predictive models
Data Type	Structured only	Structured + Unstructured
Techniques	Querying, reporting	ML, Statistics, AI
Complexity	Less complex	More complex
Goal	Business reporting	Business prediction
Example	Last year sales report	Predict next month sales



Thank You!



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